

Conic Conformal Geometric Algebra: Foundation for Two-Dimensional Conic Ge- ometry

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Abstract. We present Conic Conformal Geometric Algebra (CCGA), the geometric algebra $\mathbb{R}_{5,3}$ for the geometry of planar conics, as a systematic foundation for two-dimensional conic geometry. CCGA renames and consolidates the algebra previously introduced as QC2GA and proven isomorphic to the Geometric Algebra for Conics (GAC). We give the complete catalogue of geometric objects in both outer-product null space (OPNS) and inner-product null space (IPNS) form, with their grades and reality conditions; the join, meet, and dual operations; closed-form, geometric-algebra-native extraction of every classical conic feature directly from the dual conic vector; the versor transformation group together with a precise account of which planar maps it realises; and a unified treatment of conic intersection based on a shared-factor reduction of the regressive product. Every construction is given in closed geometric-algebra form with its reality conditions.

Keywords. Conics, Geometric Algebra, Clifford Algebra, Conic Conformal Geometric Algebra (CCGA), Geometric Algebra for Conics (GAC), Projective Geometry, OPNS/IPNS duality, Conic intersection.

1. Introduction

Geometric Algebra (GA), also known as Clifford algebra, provides a unified language for geometry in which objects and the transformations acting on them live in a single associative algebra [8, 19]. Its conformal model represents points as null vectors, circles and lines as blades, and conformal maps as versors acting by the sandwich product — a formulation that has proven fruitful in classical mechanics and physics [16, 15, 13], in computer graphics [6], and more recently in geometric deep learning [3, 21]. For the Euclidean plane,

however, the conformal model stops at circles and lines: the general conic — ellipse, parabola, hyperbola, and their degenerate limits — is outside its reach, even though conics are the natural next family of planar curves in vision, graphics, and kinematics.

Several algebras have been proposed to close this gap. Perwass embedded plane conics in $\mathbb{R}_{5,3}$ via a Veronese-type point map as a proof of concept [19]; Easter and Hitzer’s Double Conformal Geometric Algebra [9] and Goldman and Mann’s $\mathbb{R}_{4,4}$ framework [12] capture wider quadric families at the price of losing some operations; Breuils et al. introduced Quadric Conformal Geometric Algebra (QCGA, $\mathbb{R}_{9,6}$) for three-dimensional quadrics [2, 1], whose two-dimensional restriction QC2GA is again $\mathbb{R}_{5,3}$; and Hrdina, Návrát, and Vašík proposed the Geometric Algebra for Conics (GAC) [17], since used for conic fitting [18] and for conic intersection in special configurations [4]. QC2GA and GAC were recently proven isomorphic — the same algebra in two bases [5]. Each of these contributions, however, introduced its own basis, normalisation, and name, and each developed only the slice of the theory its application required: the object catalogue, the feature-extraction formulas, the transformation group, and the intersection theory have never been assembled in one place, in one notation.

This paper provides that consolidation. We adopt the name *Conic Conformal Geometric Algebra* (CCGA) for $\mathbb{R}_{5,3}$ equipped with the null working basis and point embedding of Sections 2 and 3, and we develop the algebra as a self-contained foundation: the complete taxonomy of its objects in both OPNS and IPNS form, organised by a single construction ladder; closed-form extraction of every classical conic feature directly from the dual conic vector by inner products with a fixed set of special blades; a precise characterisation of which planar maps the versor group realises; and a unified treatment of conic intersection by grade arithmetic of the regressive product. Throughout, a single design principle is enforced: *every structural operation is geometric algebra; the only numeric residue is root extraction* (square roots and polynomial roots, which carry the irreducible algebraic content of intersecting curves).

1.1. Contributions

The main contributions of this paper are:

- A self-contained construction of **Conic Conformal Geometric Algebra** (CCGA, $\mathbb{R}_{5,3}$), unifying the former QC2GA and GAC under one name and one notation (Sections 2 and 3).
- The **complete object taxonomy** of CCGA in both OPNS and IPNS form, with grades, geometric meaning, and reality conditions, organised by a single construction ladder (Sections 4 and 5).
- **Geometric-algebra-native feature extraction**: closed-form formulas reading the type, center, axes, foci, eccentricity, asymptotic directions, and tangent/polar lines of a conic directly from its dual grade-1 vector by inner products (Section 7).
- A precise characterisation of the **versor transformation group** of CCGA — which planar maps are sandwich versors (similarities) and which are

only linear outermorphisms (shears, non-uniform scalings, projective maps) (Section 8).

- A unified **intersection framework** based on a shared-factor / Bézout reduction of the regressive product, complementing the pencil method of [5] (Section 9).

1.2. Notation and Conventions

We follow the GA conventions of [8, 19]; Table 1 fixes the symbols used throughout. Throughout, $A \equiv B$ means A equals B up to a nonzero scalar, $\langle A \rangle_k$ is the grade- k part, and the dual is fixed as $A^* = A I^{-1}$ (right multiplication by the inverse pseudoscalar).

TABLE 1. Notation.

Symbol	Meaning	Convention
$A \wedge B$	outer product (join, span)	grades add
$A \cdot B$	inner product	symmetric, vectors
$A \lrcorner B$	left contraction	[7]
$A \vee B$	meet (regressive product)	$(A^* \wedge B^*)^*$
A^*	dual	$A I^{-1}$
$\tilde{A}$	reverse	
$\langle A \rangle_k$	grade- k part	
$A \equiv B$	equality up to a nonzero scalar	projective equality

Lower-case bold letters denote basis blades and multivectors (**a**); italic lower-case letters denote their scalar components ($a_1, x, y^2, \dots$); the superscript star $\mathbf{x}^*$ is the dual; and the subscript ϵ in $\mathbf{x}_\epsilon$ extracts the Euclidean (e_1, e_2) part of a vector. The contraction and outer products take priority over the full geometric product, e.g. $\mathbf{a} \wedge \mathbf{b} \mathbf{I} = (\mathbf{a} \wedge \mathbf{b}) \mathbf{I}$.

2. The CCGA Algebra $\mathbb{R}_{5,3}$

CCGA is the geometric algebra $\mathbb{R}_{5,3}$ (five positive, three negative, no null basis axes), of dimension $2^8 = 256$. It is the conic algebra first introduced by Perwass [19], developed as the two-dimensional restriction QC2GA of Quadric Conformal Geometric Algebra [1] and shown isomorphic to the Geometric Algebra for Conics (GAC) [17]; we call it Conic Conformal Geometric Algebra and relate the two bases in Section 2.6. Because the metric is *non-degenerate*, the pseudoscalar is invertible, so the dual, meet, and join are all clean — in contrast with the degenerate metric of PGA, where these require special care.

2.1. Diagonal and null bases

The orthogonal basis splits into the Euclidean pair $\{e_1, e_2\}$ (+1), three positive vectors $\{e_{+1}, e_{+2}, e_{+3}\}$ (+1), and three negative vectors $\{e_{-1}, e_{-2}, e_{-3}\}$ (−1).

The null working basis is, for $1 \leq i \leq 3$,

$$e_{o_i} = e_{+i} + e_{-i}, \quad e_{\infty_i} = \frac{1}{2}(e_{-i} - e_{+i}), \quad (1)$$

with inverse $e_{+i} = \frac{1}{2}e_{o_i} - e_{\infty_i}$, $e_{-i} = \frac{1}{2}e_{o_i} + e_{\infty_i}$. Each pair (e_{o_i}, e_{∞_i}) is null, $e_{o_i}^2 = e_{\infty_i}^2 = 0$, with hyperbolic pairing $e_{o_i} \cdot e_{\infty_i} = -1$, and the three pairs are mutually orthogonal. The resulting Gram matrix is

	e_1	e_2	e_{o_1}	e_{∞_1}	e_{o_2}	e_{∞_2}	e_{o_3}	e_{∞_3}
e_1	1	.	.	.	.	.	.	.
e_2	.	1	.	.	.	.	.	.
e_{o_1}	.	.	0	-1	.	.	.	.
e_{∞_1}	.	.	-1	0	.	.	.	.
e_{o_2}	.	.	.	.	0	-1	.	.
e_{∞_2}	.	.	.	.	-1	0	.	.
e_{o_3}	.	.	.	.	.	.	0	-1
e_{∞_3}	.	.	.	.	.	.	-1	0

(2)

This Gram matrix follows directly from the null-basis definitions (1).

Remark 2.1. A symmetric normalisation $e_{o_i} = \frac{1}{\sqrt{2}}(e_{-i} - e_{+i})$, $e_{\infty_i} = \frac{1}{\sqrt{2}}(e_{+i} + e_{-i})$ yields the same metric of Equation (2) and is convenient for deriving rotation/translation/scaling versors. We adopt the asymmetric convention (1) throughout, as it makes the point embedding (8) carry the conic coefficients directly; the two differ only by a fixed rescaling of each null pair.

2.2. Special blades and the pseudoscalar

The following blades structure every later construction:

$$e_o = e_{o_1} + e_{o_2}, \quad e_\infty = \frac{1}{2}(e_{\infty_1} + e_{\infty_2}), \quad (3a)$$

$$\bar{e}_o = e_{o_1} - e_{o_2}, \quad \bar{e}_\infty = \frac{1}{2}(e_{\infty_1} - e_{\infty_2}), \quad (3b)$$

$$I_o^\flat = \bar{e}_o \wedge e_{o_3}, \quad I_\infty^\flat = (e_{\infty_1} - e_{\infty_2}) \wedge e_{\infty_3}, \quad (3c)$$

$$I_o = e_{o_1} \wedge e_{o_2} \wedge e_{o_3}, \quad I_\infty = e_{\infty_1} \wedge e_{\infty_2} \wedge e_{\infty_3}, \quad (3d)$$

$$I_\epsilon = e_1 \wedge e_2, \quad I = I_\epsilon \wedge I_\infty \wedge I_o. \quad (3e)$$

Table 2 summarises these blades, their grades, and their geometric role.

TABLE 2. The special blades of CCGA, with grades and geometric role (Section 2.3).

Symbol	Definition	Grade	Role
e_o	$e_{o_1} + e_{o_2}$	1	combined origin
e_∞	$\frac{1}{2}(e_{\infty_1} + e_{\infty_2})$	1	isotropic infinity
$\bar{e}_o$	$e_{o_1} - e_{o_2}$	1	differential origin
$\bar{e}_\infty$	$\frac{1}{2}(e_{\infty_1} - e_{\infty_2})$	1	differential infinity
$I_o^\triangleright$	$\bar{e}_o \wedge e_{o_3}$	2	origin gauge
$I_\infty^\triangleright$	$(e_{\infty_1} - e_{\infty_2}) \wedge e_{\infty_3}$	2	infinity gauge (circular points)
I_o	$e_{o_1} \wedge e_{o_2} \wedge e_{o_3}$	3	origin 3-blade
I_∞	$e_{\infty_1} \wedge e_{\infty_2} \wedge e_{\infty_3}$	3	line/conic at infinity
I_ϵ	$e_1 \wedge e_2$	2	Euclidean pseudoscalar
I	$I_\epsilon \wedge I_\infty \wedge I_o$	8	pseudoscalar

Proposition 2.2. *The grade-1 special blades satisfy*

$$e_o \cdot e_\infty = -1, \quad \bar{e}_o \cdot \bar{e}_\infty = -1, \quad e_{o_3} \cdot e_{\infty_3} = -1, \quad (4)$$

and the two pairs are mutually orthogonal, $\{e_o, e_\infty\} \perp \{\bar{e}_o, \bar{e}_\infty\}$. The pseudoscalar squares to $I^2 = -1$, hence $I^{-1} = -I$ and $II^{-1} = 1$.

These relations follow by direct computation from Equation (2).

2.3. Geometric meaning of the special blades

The eight null directions group into the three conjugate pairs of Equation (2), each carrying a distinct piece of conic structure. On the *origin* side, $e_o = e_{o_1} + e_{o_2}$ is the combined origin, $\bar{e}_o = e_{o_1} - e_{o_2}$ the *differential* origin, and e_{o_3} the *cross-term* origin; the *infinity* side mirrors them with the isotropic e_∞ , the differential $\bar{e}_\infty$, and the cross-term e_{∞_3} . The three pairs govern, respectively, the *size*/isotropy of a round object (the radius, carried by e_∞), the $x^2 - y^2$ *anisotropy* that distinguishes the two axes (carried by $\bar{e}_o, \bar{e}_\infty$), and the *xy tilt* (carried by e_{o_3}, e_{∞_3}).

This split is the algebraic form of the principle that *shape lives on the origin side and size on the infinity side*: in the dual conic vector (16), the quadratic coefficients A, B, C sit on $e_{o_1}, e_{o_2}, e_{o_3}$ while the constant F sits on $e_{\infty_1} + e_{\infty_2}$. The infinity-gauge blade $I_\infty^\triangleright = (e_{\infty_1} - e_{\infty_2}) \wedge e_{\infty_3}$ is the pair of *circular points* I, J at infinity through which every circle passes; it controls the round sub-family and the intersection reductions of Section 9.

2.4. Subspace structure

The eight-dimensional space splits as $\mathbb{R}^8 = V_6 \oplus W_2$, where

$$V_6 = \text{span}\{e_o, e_1, e_2, e_{\infty_1}, e_{\infty_2}, e_{\infty_3}\}, \quad W_2 = \text{span}\{\bar{e}_o, e_{o_3}\}. \quad (5)$$

V_6 is the *point/conic space* — it carries the six Veronese coordinates $(1, x, y, x^2, y^2, xy)$ of a conic — while W_2 is the two-dimensional *gauge plane*, whose top blade is

exactly $I_o^\flat = \bar{e}_o \wedge e_{o_3}$. This is why $I_o^\flat$ is *the* gauge blade used to fix the outer-product conic to grade 7 (Section 4): it supplies precisely the two directions that the points cannot reach. The proof that the point embedding (8) lands in V_6 (and never in W_2) is given in Section 3.

2.5. Reciprocity identities

The two gauge blades $I_o^\flat$ and $I_\infty^\flat$ are reciprocal, and the line at infinity factors through $I_\infty^\flat$:

$$I_\infty^\flat \wedge e_\infty = -I_\infty, \quad e_{\infty_3} \wedge \bar{e}_\infty = -\frac{1}{2} I_\infty^\flat, \quad I_\infty^\flat \lrcorner I_o^\flat = -2. \quad (6)$$

The last identity makes the contraction $\lrcorner I_\infty^\flat$ the inverse of the wedge $\wedge I_o^\flat$, which is the algebraic mechanism behind the conic-intersection pipeline of Section 9; the first lets a line, as a degenerate conic, carry the whole line at infinity I_∞ .

2.6. Relation to GAC

CCGA and the Geometric Algebra for Conics (GAC) [17] are the same algebra in different bases. Writing the GAC basis $(e_1, e_2, \bar{n}_+, n_+, \bar{n}_-, n_-, \bar{n}_\times, n_\times)$ in terms of the CCGA null basis,

$$\bar{n}_+ = e_o, \quad n_+ = e_\infty, \quad \bar{n}_- = \bar{e}_o, \quad n_- = \bar{e}_\infty, \quad \bar{n}_\times = e_{o_3}, \quad n_\times = e_{\infty_3}, \quad (7)$$

exhibits the isomorphism as a pure change of basis; the full equivalence is established in [5]. The GAC names match the geometric roles of Section 2.3: the $+$ pair is isotropic (size), the $-$ pair is the $x^2 - y^2$ anisotropy, and the $\times$ pair is the xy cross term.

3. The Point Embedding

The conic point map $\mathbb{R}^2 \rightarrow \mathbb{R}^8$ is the Veronese-type embedding

$$p = e_o + p_\epsilon + \frac{1}{2} (x^2 e_{\infty_1} + y^2 e_{\infty_2}) + xy e_{\infty_3}, \quad p_\epsilon = x e_1 + y e_2. \quad (8)$$

Note that the null vector e_{o_3} does not appear in a point: this preserves the C2GA property that the inner product of two points is (minus half) their squared Euclidean distance. For p_1, p_2 at $(x_1, y_1), (x_2, y_2)$,

$$\begin{aligned} p_1 \cdot p_2 &= -\frac{x_1^2}{2} - \frac{x_2^2}{2} - \frac{y_1^2}{2} - \frac{y_2^2}{2} + x_1 x_2 + y_1 y_2 \\ &= -\frac{1}{2} ((x_1 - x_2)^2 + (y_1 - y_2)^2) = -\frac{1}{2} \text{Dist}^2(p_1, p_2), \end{aligned} \quad (9)$$

so in particular $p^2 = 0$ (every point is null). Finite points are normalised by $p \cdot e_\infty = -1$; rescaling a round object by $-(\cdot) \cdot e_\infty$ restores this canonical scale.

Points at infinity. Sending $p(t\mathbf{v})$ to infinity along a direction $\mathbf{v} = (v_x, v_y)$ and keeping the leading (degree-2) term defines the *ideal point*

$$v_\infty(\mathbf{v}) = \lim_{t \rightarrow \infty} \frac{p(t\mathbf{v})}{t^2} = \frac{v_x^2}{2} e_{\infty_1} + \frac{v_y^2}{2} e_{\infty_2} + v_x v_y e_{\infty_3}, \quad (10)$$

a null vector in I_∞ , defined up to scale and independent of the sign of $\mathbf{v}$ (directions at infinity are unoriented). The ideal points sweep the Veronese conic inside the plane at infinity; the two *circular points* are the complex pair $I = v_\infty(1, i)$, $J = v_\infty(1, -i)$, whose real bivector span is the infinity gauge,

$$I \wedge J = -i I_\infty^\flat, \quad (11)$$

so that “containing the factor $I_\infty^\flat$ ” and “passing through the circular points” are the same condition — the classical characterisation of circles [20]. We write $v'_\infty(\mathbf{v})$ for the tangent direction to the Veronese conic at $v_\infty(\mathbf{v})$; the pair $v_\infty \wedge v'_\infty$ encodes a *double* (tangential) contact with the line at infinity and is the infinity gauge of the parabola in Section 4.3.

4. Conics: OPNS and IPNS

4.1. The grade-1 IPNS conic

The fundamental fact of CCGA is that a single grade-1 vector cuts out a general conic by its inner-product null space. For an arbitrary vector $s = s_{o_1} e_{o_1} + s_{o_2} e_{o_2} + s_{o_3} e_{o_3} + s_{e_1} e_1 + s_{e_2} e_2 + s_{\infty_1} e_{\infty_1} + s_{\infty_2} e_{\infty_2} + s_{\infty_3} e_{\infty_3}$ and a point $p = p(x, y)$ of Equation (8), expanding $p \cdot s$ with the Gram matrix (2) gives

$$p \cdot s = 0 \iff Ax^2 + By^2 + Cxy + Dx + Ey + F = 0, \quad (12)$$

with the coefficient dictionary

$$A = -\frac{s_{o_1}}{2}, \quad B = -\frac{s_{o_2}}{2}, \quad C = -s_{o_3}, \quad D = s_{e_1}, \quad E = s_{e_2}, \quad F = -(s_{\infty_1} + s_{\infty_2}). \quad (13)$$

Two directions are *gauge-inert*: $p \cdot s$ never depends on s_{∞_3} , and depends on $s_{\infty_1}, s_{\infty_2}$ only through their sum — i.e. adding any multiple of e_{∞_3} or of $\bar{e}_\infty$ to s does not move the locus. The canonical (gauge-fixed) dual conic therefore lives in the six-dimensional space $\text{span}\{e_1, e_2, e_{o_1}, e_{o_2}, e_{o_3}, e_\infty\}$, matching the six projective coefficients $(A : B : C : D : E : F)$, and exhibits the principle of Section 2.3: shape (A, B, C) on the origin side, size F on the infinity side.

4.2. The grade-7 OPNS conic and the $I_o^\flat$ gauge

$$\mathbf{c} = I_o^\flat \wedge p_1 \wedge \cdots \wedge p_5, \quad \mathbf{c}^* = \mathbf{c} I^{-1} \text{ is the grade-1 conic vector.} \quad (14)$$

Expanding the five-point wedge and collecting the seven blade components — two of which share a coefficient — gives the closed blade form of $\mathbf{c}$ whose

dual is the grade-1 conic vector $\mathbf{c}^*$ of Equation (12):

$$\mathbf{c} = 2A(I_\epsilon \wedge e_{\infty_2} \wedge e_{\infty_3} \wedge I_o) + 2B(I_\epsilon \wedge e_{\infty_1} \wedge e_{\infty_3} \wedge I_o) + C(I_\epsilon \wedge e_{\infty_1} \wedge e_{\infty_2} \wedge I_o) \\ - D(e_2 \wedge I_\infty \wedge I_o) - E(e_1 \wedge I_\infty \wedge I_o) - \frac{1}{2}F(I_\epsilon \wedge I_\infty \wedge e_o) = \mathbf{c}^* I, \quad (15)$$

$$\mathbf{c}^* = -(2A e_{o_1} + 2B e_{o_2} + C e_{o_3}) + D e_1 + E e_2 - F e_\infty. \quad (16)$$

The reading $\mathbf{p} \cdot \mathbf{c}^* = 0 \iff \mathbf{p} \wedge \mathbf{c} = 0$ recovers the locus (12), so the six conic coefficients sit directly in the grade-1 vector (16).

4.3. Axis-aligned conics by infinity gauge

An axis-aligned conic $Ax^2 + By^2 + Dx + Ey + F = 0$ has no cross term, i.e. $C = 0$, leaving five coefficients and four degrees of freedom; it can therefore be built from *four* points. The cross term is removed by wedging with e_{∞_3} , which annihilates the xy carrier of every point:

$$\mathbf{p} \wedge e_{\infty_3} = (e_o + \mathbf{p}_\epsilon + \frac{1}{2}(x^2 e_{\infty_1} + y^2 e_{\infty_2})) \wedge e_{\infty_3}. \quad (17)$$

Hence an axis-aligned conic is a general conic with four points sent to the e_{∞_3} direction:

$$\mathbf{c} = I_o^\triangleright \wedge \mathbf{p}_1 \wedge \mathbf{p}_2 \wedge \mathbf{p}_3 \wedge \mathbf{p}_4 \wedge e_{\infty_3}. \quad (18)$$

Equivalently, $C = 0$ is the statement that the e_{o_3} coefficient of $\mathbf{c}^*$ in (16) vanishes.

The infinity gauge. The four points in (18) carry one redundant degree of freedom: the *type* of a conic is decided entirely by its two points at infinity — its intersection with the line at infinity I_∞ . Spending that degree of freedom, by replacing the fourth point with a fixed grade-2 *infinity gauge* blade B_∞ , produces an axis-aligned conic of prescribed type from only three points:

$$\mathbf{c} = \mathbf{p}_1 \wedge \mathbf{p}_2 \wedge \mathbf{p}_3 \wedge B_\infty \wedge I_o^\triangleright, \quad B_\infty = (\alpha e_{\infty_1} + \beta e_{\infty_2}) \wedge e_{\infty_3}. \quad (19)$$

B_∞ is the pair of ideal points of $\mathbf{c}$, the two solutions of its quadratic part on I_∞ ; choosing $B_\infty = \mathbf{p}_4 \wedge e_{\infty_3}$ recovers the general axis-aligned conic (18). The smooth types are listed in Table 3.

TABLE 3. Axis-aligned conics by infinity gauge B_∞ (smooth types).

Type	B_∞	ideal points	Δ_2
circle	$(e_{\infty_1} - e_{\infty_2}) \wedge e_{\infty_3} = I_\infty^\triangleright$	imaginary $(1, \pm i)$	$> 0, A=B$
ellipse $a:b$	$(a^2 e_{\infty_1} - b^2 e_{\infty_2}) \wedge e_{\infty_3}$	imaginary $(a, \pm ib)$	> 0
hyperbola $a:b$	$(a^2 e_{\infty_1} + b^2 e_{\infty_2}) \wedge e_{\infty_3}$	real $(a, \pm b)$	< 0
parabola $\parallel x$	$e_{\infty_1} \wedge e_{\infty_3}$	real double $(1, 0)$	0
parabola $\parallel y$	$e_{\infty_2} \wedge e_{\infty_3}$	real double $(0, 1)$	0

Proposition 4.1 (type from the infinity gauge). *The conic (19) has $C = 0$ and quadratic coefficients $A \propto \alpha$, $B \propto -\beta$, hence discriminant $\Delta_2 = AB \propto -\alpha\beta$. Therefore*

$$\begin{aligned} \alpha\beta < 0 &\Rightarrow \Delta_2 > 0 \text{ (ellipse)}, & \alpha\beta = 0 &\Rightarrow \Delta_2 = 0 \text{ (parabola)}, \\ \alpha\beta > 0 &\Rightarrow \Delta_2 < 0 \text{ (hyperbola)}, \end{aligned} \quad (20)$$

which is the projective classification by incidence with I_∞ : an imaginary, double, or real ideal pair (the discriminants Δ_2, Δ_3 are defined in Section 7).

The ideal pairs are wedges of the at-infinity points of Section 3: the real pair $v_\infty(a, b) \wedge v_\infty(a, -b) \equiv (a^2 e_{\infty_1} + b^2 e_{\infty_2}) \wedge e_{\infty_3}$ (hyperbola), its imaginary counterpart $(a^2 e_{\infty_1} - b^2 e_{\infty_2}) \wedge e_{\infty_3}$ (ellipse), and the double contact $v_\infty(1, 0) \wedge v'_\infty(1, 0) \equiv e_{\infty_1} \wedge e_{\infty_3}$ (parabola). The circle is the isotropic case $\alpha = -\beta$, where $B_\infty = I_\infty^*$ is the circular-point pair of Section 2.3; the construction then specialises to the circle, and (19) to the standard axis-aligned ellipse, up to scale.

Degenerate conics. Within each Δ_2 class a degenerate member ($\Delta_3 = 0$) appears when the three points are placed in special position; B_∞ still fixes Δ_2 , while Δ_3 drops to zero:

- *Two crossing lines* (degenerate hyperbola): $B_\infty = (e_{\infty_1} + e_{\infty_2}) \wedge e_{\infty_3}$ with the three points through the common centre, e.g. yielding $x^2 - y^2 = 0$.
- *Two parallel lines* (degenerate parabola): $B_\infty = e_{\infty_2} \wedge e_{\infty_3}$ with three points on two vertical lines, e.g. $x^2 - 1 = 0$.
- *A point* (degenerate ellipse, two imaginary lines): the $r \rightarrow 0$ limit of an imaginary circle, $x^2 + y^2 = 0$; having no three distinct real points, it arises only as a limit.

Since the type test keys on Δ_2 alone, a degenerate hyperbola still reports as a hyperbola; genuineness is the separate condition $\Delta_3 \neq 0$. This places the whole family on the classical six-cell classification of Table 4: B_∞ chooses the column (Δ_2) and the point configuration chooses the row (Δ_3).

TABLE 4. The six conic types as infinity gauge (Δ_2) $\times$ point configuration (Δ_3), reproducing the classical classification [20].

	$\Delta_2 > 0$ ($\alpha\beta < 0$)	$\Delta_2 = 0$ ($\alpha\beta = 0$)	$\Delta_2 < 0$ ($\alpha\beta > 0$)
$\Delta_3 \neq 0$	ellipse / circle	parabola	hyperbola
$\Delta_3 = 0$	point	two parallel lines	two crossing lines

Tilted (non-axis-aligned) conics follow by applying a rotation versor (Section 8) to (19), or directly from off-axis ideal directions in B_∞ (general v_∞ factors). Three-dimensional quadric-surface variants are out of scope for this two-dimensional foundation and belong to the QCGA lift discussed in the conclusion.

4.4. Axis-aligned conics by dual construction

Dually, an axis-aligned conic is the grade-1 vector $s = \mathbf{c}^*$ of (16) with $C = 0$. Splitting the origin part with $e_o = e_{o_1} + e_{o_2}$ and $\bar{e}_o = e_{o_1} - e_{o_2}$, so that $-2A e_{o_1} - 2B e_{o_2} = -(A+B) e_o - (A-B) \bar{e}_o$,

$$s = -(A+B) e_o - (A-B) \bar{e}_o + D e_1 + E e_2 - F e_\infty. \quad (21)$$

A point lies on the conic iff $\mathbf{q} \cdot s = 0$ (12). For the centred conics ($D = E = 0$, $F = -1$) this gives the closed forms of Table 5; the circle is the familiar centre-point form $\mathbf{p}_c - \frac{r^2}{2} e_\infty$.

TABLE 5. Axis-aligned conics by dual construction (grade-1 IPNS vector). The ellipse/hyperbola/parabola are centred at the origin; a centre (c_x, c_y) adds $-2Ac_x e_1 - 2Bc_y e_2$ and sets $F = Ac_x^2 + Bc_y^2 - 1$.

Type	dual vector s
circle (c, r)	$\mathbf{p}_c - \frac{r^2}{2} e_\infty$
ellipse (a, b)	$-\left(\frac{1}{a^2} + \frac{1}{b^2}\right) e_o - \left(\frac{1}{a^2} - \frac{1}{b^2}\right) \bar{e}_o + e_\infty$
hyperbola (a, b)	$-\left(\frac{1}{a^2} - \frac{1}{b^2}\right) e_o - \left(\frac{1}{a^2} + \frac{1}{b^2}\right) \bar{e}_o + e_\infty$
parabola $y^2=4px$	$-e_o + \bar{e}_o - 4p e_1$
parabola $x^2=4py$	$-e_o - \bar{e}_o - 4p e_2$

The anisotropic round point. The circle's $\mathbf{p}_c - \frac{1}{2}r^2 e_\infty$ generalises by giving the point two independent weights, one per conformal origin. For weights w_1, w_2 define the *anisotropic point* at (c_x, c_y) as one independently weighted bundle per axis,

$$\mathbf{p}^{(w_1, w_2)} = w_1 \left(e_{o_1} + c_x e_1 + \frac{c_x^2}{2} e_{\infty_1} \right) + w_2 \left(e_{o_2} + c_y e_2 + \frac{c_y^2}{2} e_{\infty_2} \right) + w_1 w_2 c_x c_y e_{\infty_3}, \quad (22)$$

which is *null for every* w_1, w_2 — the per-axis pairings $e_{o_i} \cdot e_{\infty_i} = -1$ cancel axis by axis — and obeys, for a test point $\mathbf{q} = \mathbf{q}(x, y)$, the weighted distance identity $\mathbf{q} \cdot \mathbf{p}^{(w_1, w_2)} = -\frac{1}{2} [w_1(x - c_x)^2 + w_2(y - c_y)^2]$. Attaching a radius yields a *single dual vector for every axis-aligned conic*, the *anisotropic round point*

$$\sigma(c_x, c_y, r, w_1, w_2) = w_1 \left(e_{o_1} + c_x e_1 + \frac{c_x^2}{2} e_{\infty_1} \right) + w_2 \left(e_{o_2} + c_y e_2 + \frac{c_y^2}{2} e_{\infty_2} \right) + w_1 w_2 c_x c_y e_{\infty_3} - \frac{1}{2} r^2 e_\infty, \quad (23)$$

whose locus $\mathbf{q} \cdot \sigma = 0$ is the conic $w_1(x - c_x)^2 + w_2(y - c_y)^2 = r^2$, with quadratic coefficients $A = w_1$, $B = w_2$ and square $\sigma^2 = \frac{1}{2}r^2(w_1 + w_2)$. The **weight ratio** $w_1 : w_2 = 1/a^2 : 1/b^2$ **fixes the** (a, b) **shape** and r fixes the size, so (23) realises *every* axis-aligned conic: $w_1 = w_2$ gives a circle, $w_1, w_2 > 0$ an ellipse (semi-axes a, b for $w_i = 1/a_i^2$, $r = 1$), and $w_2 < 0$ a hyperbola. The sign of the radius term is the reality ($-\frac{1}{2}r^2 e_\infty$ the real conic, $+\frac{1}{2}r^2 e_\infty$

its imaginary partner), exactly as for the circle, whose ordinary round point $p_c - \frac{1}{2}r^2e_\infty$ is the isotropic case $w_1 = w_2 = 1$. Expression (23) agrees with the canonical vector of Table 5 up to the gauge-inert $\bar{e}_\infty, e_{\infty_3}$ terms, which do not move the locus.

Remark 4.2 (why CCGA, and why this is still a conic). CGA’s round point $p_c - \frac{1}{2}r^2e_\infty$ has a single weight and can carry only an isotropic radius (circles); CCGA’s *two* conformal origins e_{o_1}, e_{o_2} let the round point carry an *anisotropic* radius, i.e. an ellipse or hyperbola. The weights are exactly the conic’s quadratic coefficients $(A, B) = (w_1, w_2)$, and allowing the off-diagonal weight on e_{o_3} adds the cross term C , recovering the general (tilted) dual conic. A “fully weighted point” is therefore a quadratic form — a conic — rather than a new primitive: the isotropic point of Section 3 remains the base element of the algebra, and σ is read as a *derived* round object.

Proposition 4.3 (type from the origin balance). *In (21) the conic type is set by the balance of the combined and differential origins, $\bar{e}_o/e_o$: writing $A = 1/a^2$ and $B = \pm 1/b^2$,*

$$\begin{aligned} |A-B| < |A+B| \text{ (ellipse),} \quad & |A-B| = |A+B| \text{ (parabola),} \\ & |A-B| > |A+B| \text{ (hyperbola),} \end{aligned} \quad (24)$$

the circle being the isotropic limit $\bar{e}_o = 0$ ($A = B$), and the parabola the case where one of A, B vanishes.

This is the dual of Proposition 4.1: the wedge construction fixes the type by the *infinity* gauge B_∞ (the conic’s ideal points), whereas the dual construction fixes it by the *origin* blades $e_o, \bar{e}_o$ (its shape) — the gauge duality “shape on the origin side, size on the infinity side” of Section 2.3. The circle’s clean centre-point form $p_c - \frac{r^2}{2}e_\infty$ is exactly the isotropic case where the $\bar{e}_o$ term drops out, while the “ $\pm$ ” that distinguishes ellipse from hyperbola is the sign carried by the $\bar{e}_o$ anisotropy term. Off-centre conics follow by a translator versor (Section 8), and the degenerate members ($\Delta_3 = 0$) are the same special-position limits as on the wedge side.

5. The Object Taxonomy

5.1. The construction ladder

Every CCGA object is a wedge of points with at most the two gauge blades $I_o^\triangleright$ (origin) and $I_\infty^\triangleright$ (infinity, the circular-point pair). Three ladders organise the whole zoo, distinguished by which gauge they carry.

1. **Bare multipole ladder** (no gauge): the plain wedge of k points, $p_1 \wedge \cdots \wedge p_k$, grade k for $k = 1, \dots, 5$ (point, twopole, tripole, quadpole, pentapole). Five points already fix a conic as a point-locus; the pentapole is the last rung.
2. **$I_o^\triangleright$ -gauged conic ladder**: wedging $I_o^\triangleright$ promotes a blade by two grades and gauge-fixes it so the dual is clean (Section 4). For $n \leq 4$ points

$p_1 \wedge \cdots \wedge p_n \wedge I_o^\triangleright$ (grade $n + 2$) is a *pencil* — the gauge-fixed family of conics through the n points; the $n = 2$ rung is the *gauged dipole* (the conic-gauge point pair, the natural conic $\vee$ line result) and the $n = 4$ rung is the conic $\vee$ conic intersection blade. A *full CCGA conic is precisely a grade-7 blade ending in $\wedge I_o^\triangleright$* . The general conic is $p_1 \wedge \cdots \wedge p_5 \wedge I_o^\triangleright$; the smooth named conics are *three points + an ideal-point pair + $I_o^\triangleright$* , the pair being two real ideal points (hyperbola, $\Delta_2 > 0$), one double ideal point (parabola, tangent to I_∞ , $\Delta_2 = 0$), or two imaginary ideal points (ellipse, $\Delta_2 < 0$). When that pair is the circular points $I_\infty^\triangleright$ the ellipse becomes a circle — which is therefore a CGA object (next ladder), not listed here.

3. **CGA embedded in CCGA** (the $I_\infty^\triangleright$ sub-array): wedging the circular-point pair $I_\infty^\triangleright$ embeds the standard CGA round/flat family inside CCGA at grade = (CGA grade) + 2. Round point $p \wedge I_\infty^\triangleright$, point pair $p_1 \wedge p_2 \wedge I_\infty^\triangleright$, circle $p_1 \wedge p_2 \wedge p_3 \wedge I_\infty^\triangleright$, with the flat versions (flat point, line) carrying an extra e_∞ . These are CGA objects, not full CCGA conics.

5.2. Master object table

TABLE 6. The CCGA object zoo, sorted by OPNS grade, in three construction blocks: the bare multipole ladder, the $I_o^\flat$ -gauged conic ladder, and the $I_\infty^\flat$ CGA-embedded sub-array; followed by flats / ideal elements. The ideal-point pair of a smooth conic is written $v_\infty^{(1)}, v_\infty^{(2)}$ (real/imaginary) or v_∞, v'_∞ (double); “CGA gr.” is the grade inside the CGA sub-algebra (CCGA grade -2).

Object	OPNS form	gr(OPNS)	gr(IPNS)	Reality
<i>Bare multipole ladder (points only)</i>				
Point	p	1	7	null
Twopole	$p_1 \wedge p_2$	2	6	$\text{sign}(\cdot)^2$
Tripole	$p_1 \wedge p_2 \wedge p_3$	3	5	—
Quadpole	$p_1 \wedge \cdots \wedge p_4$	4	4	—
Pentapole	$p_1 \wedge \cdots \wedge p_5$	5	3	—
<i>$I_o^\flat$-gauged conic ladder (a CCGA conic ends in $\wedge I_o^\flat$)</i>				
Pencil ($n \leq 4$)	$p_1 \wedge \cdots \wedge p_n \wedge I_o^\flat$	$n+2$	$6-n$	—
Conic (general)	$p_1 \wedge \cdots \wedge p_5 \wedge I_o^\flat$	7	1	Δ_2, Δ_3
Hyperbola	$p_1 \wedge p_2 \wedge p_3 \wedge v_\infty^{(1)} \wedge v_\infty^{(2)} \wedge I_o^\flat$	7	1	$\Delta_2 > 0$
Parabola	$p_1 \wedge p_2 \wedge p_3 \wedge v_\infty \wedge v'_\infty \wedge I_o^\flat$	7	1	$\Delta_2 = 0$
Ellipse	$p_1 \wedge p_2 \wedge p_3 \wedge v_\infty^{(1)} \wedge v_\infty^{(2)} \wedge I_o^\flat$	7	1	$\Delta_2 < 0$
<i>CGA embedded in CCGA (the $I_\infty^\flat$ sub-array)</i>				
Round point	$p \wedge I_\infty^\flat$	3	—	CGA gr. 1
Point pair	$p_1 \wedge p_2 \wedge I_\infty^\flat$	4	—	CGA gr. 2
Flat point (CGA)	$p \wedge e_\infty \wedge I_\infty^\flat$	4	—	CGA gr. 2
Circle (CGA)	$p_1 \wedge p_2 \wedge p_3 \wedge I_\infty^\flat$	5	—	CGA gr. 3
Line (CGA)	$p_1 \wedge p_2 \wedge e_\infty \wedge I_\infty^\flat$	5	—	CGA gr. 3
<i>Flats / ideal elements</i>				
Ideal point	(drop e_o part)	1	—	n/a
Line at infinity	I_∞	3	—	n/a
Flat point	$p \wedge I_\infty$	4	—	n/a
Conic at infinity	$I_o^\flat \wedge I_\infty$	5	—	n/a
Pseudoscalar	I	8	0	—

The ideal points $v_\infty = v_\infty(v_x, v_y) \in I_\infty$ are the asymptotic directions the conic meets on the line at infinity; the ellipse uses the imaginary pair encoded by the real bivector $(a^2 e_{\infty_1} - b^2 e_{\infty_2}) \wedge e_{\infty_3}$ (a weighted $I_\infty^\flat$, which for $a = b$ degenerates to the circular points and hence to the CGA circle).

5.3. Reality conditions

Wedges of real points always produce real objects, but the constructions of Table 6 extend by linearity to multivectors with no real locus (imaginary circles, conjugate point pairs), and the algebra detects this by a sign — always

read from a *square*, never from a raw coefficient, so that the test is invariant under rescaling.

- **Points** are null, $p^2 = 0$ (9). A *round point* (point with radius) $p_c \mp \frac{r^2}{2}e_\infty$ squares to $\pm r^2$: positive for a real circle, negative for its imaginary partner. The scale-invariant radius of any round grade-1 object is

$$r^2 = \frac{s^2}{(s \cdot e_\infty)^2}, \quad (25)$$

so sign s^2 separates *real* (> 0), *degenerate / tangent* ($= 0$), and *imaginary* (< 0). Ideal points have $s \cdot e_\infty = 0$: they carry no radius, and reality at infinity is read from the sign of the quadratic form (Section 7).

- **Twopoles** $P = p_1 \wedge p_2$ carry their reality in P^2 : positive for two real points, zero for a coincident (tangent) pair, negative for a complex-conjugate pair. The same test governs the CGA point pair $p_1 \wedge p_2 \wedge I_\infty^\flat$ and the dipoles extracted in Section 9.
- **Conics** use the pair of discriminants (Δ_2, Δ_3) of Section 7: $\Delta_3 \neq 0$ selects the smooth types (ellipse/parabola/hyperbola by the sign of Δ_2 , as in Table 4), $\Delta_3 = 0$ the degenerate ones; within the ellipse class, real versus imaginary is the sign of s^2 as for rounds.

6. Operations: Join, Meet, and Dual

Because the metric of $\mathbb{R}_{5,3}$ is non-degenerate, the pseudoscalar is invertible (Proposition 2.2) and the three incidence operations are defined for every pair of blades, with one orientation convention fixed once and used everywhere:

$$A^* = A I^{-1}, \quad (A^*) I = A, \quad A \vee B = (A^* \wedge B^*)^*. \quad (26)$$

The *join* $\wedge$ spans: it is the construction ladder of Section 5, and grades add when the factors share no common subspace. The *dual* exchanges the two null-space representations of the same object,

$$q \wedge \mathbf{c} = 0 \iff q \cdot \mathbf{c}^* = 0, \quad (27)$$

the outer-product (OPNS) and inner-product (IPNS) incidence tests of Equations (12) and (14). The *meet* $\vee$ intersects.

Grade arithmetic. For blades in general position the meet obeys

$$\text{gr}(A \vee B) = \text{gr}(A) + \text{gr}(B) - 8, \quad (28)$$

so a meet is geometrically non-trivial only when the grades sum to at least 8. At exactly 8 the meet collapses to a scalar, which is precisely the incidence test: for a conic $\mathbf{c}$ (grade 7) and a point q (grade 1), $q \vee \mathbf{c} = 0$ iff q lies on $\mathbf{c}$, in agreement with (27). Below the threshold the regressive product vanishes identically: two low-grade point-objects do not *meet* — they *join*. This asymmetry (join builds, meet intersects, meet needs grade-sum ≥ 8) organises the whole interaction calculus of Section 9, where the meaningful meets of the grade-7 conic are tabulated.

Gauge stripping. The conic constructions of Section 5 end in $\wedge I_o^\flat$; their meets therefore carry an $I_o^\flat$ factor that must be removed before the point content can be read. The reciprocity identity $I_\infty^\flat \lrcorner I_o^\flat = -2$ of Equation (6) makes the contraction $X \mapsto I_\infty^\flat \lrcorner X$ invert the wedge $X \mapsto X \wedge I_o^\flat$ on such blades: wedging with the origin gauge is undone by contracting with the infinity gauge. This single mechanism powers the quadpole recovery of Section 9 and the centre extraction of Section 7.

7. Geometric-Algebra-Native Feature Extraction

Every classical feature of a conic — type, centre, axes, semi-axes, eccentricity, foci, asymptotic directions, tangent and polar lines — can be read directly off the dual vector $s = \mathbf{c}^*$ by inner products with the special blades of Table 2 and meets, with no coordinate matrix, no eigensolver, and no linear system. The only steps that leave the algebra are $\sqrt{}$ and atan2 , the irreducible scalar content of axis lengths and angles.

7.1. Coefficients and the trace–deviatoric split

Pairing the canonical dual vector (16) with the Gram matrix (2) inverts the coefficient dictionary (13):

$$A = \frac{1}{2} s e_{\infty 1}, \quad B = \frac{1}{2} s e_{\infty 2}, \quad C = s e_{\infty 3}, \quad D = s e_1, \quad E = s e_2, \quad F = s e_o. \quad (29)$$

The quadratic part splits along the conjugate pairs of Section 2.3 into an *isotropic* scalar and a *deviatoric* spin-2 pair:

$$\underbrace{A + B = s \cdot e_\infty}_{\text{trace (size)}}, \quad \underbrace{(A - B, C) = (s \cdot \bar{e}_\infty, s \cdot e_{\infty 3})}_{\text{anisotropy; polar angle } = 2\theta_{\text{axis}}}. \quad (30)$$

The deviatoric pair is a symmetric-square (Veronese) object: its polar angle is *twice* the axis angle of the conic, which is also why the rotation versor of Section 8 acts on it with angle 2α .

7.2. Type: the discriminants as inner products and meets

The shape discriminant Δ_2 is a closed quadratic form in the three infinity pairings of (30):

$$\Delta_2 = AB - \frac{C^2}{4} = \frac{1}{4} \left[(s \cdot e_\infty)^2 - (s \cdot \bar{e}_\infty)^2 - (s \cdot e_{\infty 3})^2 \right]. \quad (31)$$

The determinant discriminant $\Delta_3 = \det M_3$ (with M_3 the symmetric coefficient matrix of the conic) is obtained as a *meet of three dual lines* [5]. The three rows of M_3 are the polar lines (cf. Section 7.6) of the two ideal basis directions and of the origin; as grade-7 blades,

$$\begin{aligned} \ell_1 &= (Ae_1 + \frac{C}{2}e_2 - \frac{D}{2}e_\infty)I, & \ell_2 &= (\frac{C}{2}e_1 + Be_2 - \frac{E}{2}e_\infty)I, \\ \ell_3 &= (\frac{D}{2}e_1 + \frac{E}{2}e_2 - Fe_\infty)I, \end{aligned} \quad (32)$$

and their meets evaluate, in closed form,

$$\ell_1 \vee \ell_2 = \frac{1}{2}(\Delta_2 e_o + x_c e_1 + y_c e_2) \wedge I_o^\flat \wedge I_\infty, \quad (33)$$

$$\ell_1 \vee \ell_2 \vee \ell_3 = \frac{1}{2} \Delta_3 I_o^\flat \wedge I_\infty, \quad (34)$$

with $x_c = \frac{CE}{4} - \frac{BD}{2}$ and $y_c = \frac{CD}{4} - \frac{AE}{2}$. The pair (Δ_2, Δ_3) then classifies the conic by Table 4: the sign of Δ_2 chooses ellipse/parabola/hyperbola, and $\Delta_3 = 0$ flags the degenerate members.

7.3. Centre

Equation (33) exhibits the centre as a geometric object: the grade-6 blade $\ell_1 \vee \ell_2$ is, up to the gauge factor $I_o^\flat \wedge I_\infty$, the *flat point* $p_c = \Delta_2 e_o + x_c e_1 + y_c e_2$ in homogeneous form — the pole of the line at infinity. For a central conic ($\Delta_2 \neq 0$) the affine centre is $(x_c/\Delta_2, y_c/\Delta_2)$; for a parabola ($\Delta_2 = 0$) the e_o component vanishes and the pole of I_∞ degenerates to an *ideal* point in the axis direction — the parabola's point of tangency with the line at infinity, which lies on the parabola itself. Equivalently, in pure inner products: with $s_i = s \cdot e_{\infty_i}$ and $\sigma_k = s \cdot e_k$,

$$c_x = \frac{s_3 \sigma_2 - s_2 \sigma_1}{s_1 s_2 - s_3^2}, \quad c_y = \frac{s_3 \sigma_1 - s_1 \sigma_2}{s_1 s_2 - s_3^2}, \quad s_1 s_2 - s_3^2 = 4\Delta_2. \quad (35)$$

7.4. Axes, semi-axes, eccentricity, foci

The eigen-decomposition of the quadratic form is already solved by the split (30) — no eigensolver is needed. The shape eigenvalues and the axis angle are

$$\lambda_\pm = \frac{1}{2} (s \cdot e_\infty) \pm \frac{1}{2} \sqrt{(s \cdot \bar{e}_\infty)^2 + (s \cdot e_{\infty_3})^2}, \quad \theta_{\text{axis}} = \frac{1}{2} \text{atan2}(s \cdot e_{\infty_3}, s \cdot \bar{e}_\infty), \quad (36)$$

and the value of the conic at its own centre is the GA incidence number $F' = p_c \cdot s$ (with p_c the embedded centre point), giving the semi-axes

$$a_i^2 = -\frac{p_c \cdot s}{\lambda_i}. \quad (37)$$

Eccentricity and foci follow classically: $c = \sqrt{|a^2 \mp b^2|}$ (ellipse $-$, hyperbola $+$), $e = c/a$, and the foci are $p_c \pm c \hat{u}_{\theta_{\text{axis}}}$ (embedded by Equation (8)); for the parabola the focus sits at focal distance from the vertex along the axis direction returned by Section 7.3. Note where the scalar boundary lies: Equations (36) and (37) use exactly one $\sqrt{}$ and one atan2 — the irreducible content of lengths and angles — while everything else is inner products.

7.5. Asymptotic directions: incidence with infinity

An ideal point (10) lies on the conic iff

$$v_\infty(\mathbf{v}) \cdot s = 0 \iff Av_x^2 + Cv_x v_y + Bv_y^2 = 0, \quad (38)$$

a binary quadratic whose root structure is the projective type classification of Proposition 4.1: two real roots give the hyperbola's asymptotic directions, one double root the parabola's axis direction, two imaginary roots the ellipse (and the circular points I, J exactly when $A = B, C = 0$). As a blade computation

this is the meet $\mathbf{c} \vee I_\infty$ — grade $7 + 3 - 8 = 2$ by Equation (28) — the conic's *ideal dipole*, whose reality is read from its square as in Section 5.3.

7.6. Tangents and polars

The polar line of a point $\mathbf{q} = (x, y)$ is assembled from the point-map differentials $\partial_x \mathbf{p} = e_1 + x e_{\infty_1} + y e_{\infty_3}$ and $\partial_y \mathbf{p} = e_2 + y e_{\infty_2} + x e_{\infty_3}$:

$$\ell_{\mathbf{q}} : \underbrace{\frac{1}{2}(\partial_x \mathbf{p} \cdot s)}_{n_x} X + \underbrace{\frac{1}{2}(\partial_y \mathbf{p} \cdot s)}_{n_y} Y + \underbrace{(e_o + \frac{x}{2}e_1 + \frac{y}{2}e_2) \cdot s}_{n_0} = 0, \quad (39)$$

encoded as the dual line $n_x e_1 + n_y e_2 - n_0 e_\infty$. This is the pole–polar pairing $M_3(x, y, 1)^\top$ in GA form: when $\mathbf{q}$ lies on the conic ($\mathbf{q} \cdot s = 0$) the polar is the *tangent* at $\mathbf{q}$; the polar of the centre is the line at infinity (consistently with Section 7.3); and the normal at a contact point is the perpendicular line through it, $-n_y X + n_x Y + (n_y x - n_x y) = 0$. The three dual lines (32) are precisely the polars of the two ideal basis directions and of the origin, closing the loop with Section 7.2.

8. Transformations as Versors

A versor V acts on every object — point, multipole, conic, pencil blade — by the same sandwich product $X \mapsto V X \tilde{V}$, and the action is an outermorphism: it distributes over $\wedge$, commutes with the dual, and therefore preserves incidence (27). This section gives the versors of the planar similarities explicitly, then delimits precisely which planar maps admit a versor at all.

8.1. The similarity versors

Because the point embedding (8) is quadratic, each Euclidean generator acts on *three* carrier pairs at once — the linear coordinates on e_1, e_2 , the squares on $e_{\infty_1}, e_{\infty_2}$, and the cross term on e_{∞_3} — and each versor below is the commuting product of the corresponding simple factors.

Translation (t_x, t_y) . $T = T_x(t_x) T_y(t_y)$ with

$$\begin{aligned} T_x(\tau) &= \left(1 - \frac{\tau}{2} e_1 \wedge e_{\infty_1}\right) \left(1 - \frac{\tau}{2} e_2 \wedge e_{\infty_3}\right), \\ T_y(\tau) &= \left(1 - \frac{\tau}{2} e_2 \wedge e_{\infty_2}\right) \left(1 - \frac{\tau}{2} e_1 \wedge e_{\infty_3}\right). \end{aligned} \quad (40)$$

The first factor of each is the familiar CGA translator on the matching conformal pair; the second updates the cross term xy , which a translation in x changes by τy .

Rotation by α . $R(\alpha) = e^{\alpha E} e^{\alpha K}$ with the commuting pair

$$E = -\frac{1}{2} e_1 \wedge e_2, \quad K = \bar{e}_o \wedge e_{\infty_3} - e_{o_3} \wedge \bar{e}_\infty, \quad K^3 = -4K, \quad (41)$$

$$e^{\alpha E} = \cos \frac{\alpha}{2} - \sin \frac{\alpha}{2} e_1 \wedge e_2, \quad e^{\alpha K} = 1 + \frac{\sin 2\alpha}{2} K + \frac{1 - \cos 2\alpha}{4} K^2. \quad (42)$$

E rotates the linear carriers at the half angle, as in CGA; K is *not* simple-exponential at the half angle — it satisfies $K^3 = -4K$ and rotates the deviatoric pair $(\bar{e}_\infty, e_{\infty_3})$ – $(\bar{e}_o, e_{o_3})$ at the *double* angle 2α ,

the symmetric-square action already met in Equation (30): a conic's anisotropy returns to itself after a half turn.

Uniform scaling by s . With $u = \frac{1}{2} \ln s$,

$$D(s) = \prod_{i=1}^3 (\cosh u + \sinh u \, e_{o_i} \wedge e_{\infty_i}), \quad (43)$$

one boost per conformal pair, all three commuting.

Reflection. Across the x -axis the versor is the grade-3 blade $V_x = e_2 \wedge e_{o_3} \wedge e_{\infty_3}$, which negates exactly the y -odd carriers (e_2 and the cross-term pair); an arbitrary line of direction angle θ through a point c reflects by the conjugate $T(c) R(\theta) V_x \tilde{R}(\theta) T(-c)$. Likewise rotations and scalings about an arbitrary centre are the translated conjugates $T(c) V T(-c)$.

8.2. Which planar maps are versors

The versors above generate, on the full conic family, exactly the *similarity group* of the plane: translations, rotations, reflections, and uniform scalings, together with their compositions. The boundary is sharp on both sides.

No versor beyond similarities. A planar map φ can act by a versor only if its lift to the Veronese coordinates $(1, x, y, \frac{x^2}{2}, \frac{y^2}{2}, xy)$ extends to an *orthogonal* map of $\mathbb{R}_{5,3}$, since sandwiching preserves the metric. A shear or a non-uniform scaling does extend to a linear outermorphism of the lift — it maps conics to conics projectively — but the induced linear map fails to preserve the inner product (it deforms the distance pairing (9)), so no versor exists: such maps act on CCGA only as outermorphisms, applied matrix-wise, with none of the versor calculus available. The same holds a fortiori for general projective maps.

Versors beyond similarities exist, but not on conics. The round (CGA) sub-family of Section 5 admits the usual conformal extensions: the invertible grade-1 vector $\sigma = p_c - \frac{r^2}{2} e_\infty$ ($\sigma^2 = r^2$) is a versor implementing inversion in the circle (p_c, r) , and $\sigma T(\mathbf{b}) \sigma$ is a transversion. These are Möbius maps: they preserve the circle/line family but send a general conic to a *quartic* curve, which has no CCGA representative. Their sandwich is still an algebra automorphism — it maps a grade-7 conic blade to a grade-7 blade — but the resulting conic is *not* the geometric image of the original curve; the versor represents the Möbius map on the round sub-family only. The transformation repertoire of CCGA on *all* conics is therefore exactly the similarities, with the conformal extension available on the CGA sub-array — consistent with GAC, whose additional conic-preserving versors [17] are likewise similarity-type on the general family.

9. Conic Intersections

Two conics meet in four points, counted with multiplicity over the complex projective plane (Bézout [11]). This section turns that statement into blade arithmetic: the regressive product delivers every intersection as a single blade

(Section 9.1), the gauge mechanism of Section 6 strips that blade to a point object (Section 9.2), and a shared-factor principle predicts, for every pair in the zoo, how many of the four points are finite and how to land them (Section 9.3).

9.1. Meet grade arithmetic

Applying Equation (28) to the grade-7 conic against the taxonomy of Table 6 gives the complete interaction row of Table 7: every meet of a conic is geometrically meaningful, ranging from the scalar incidence test up to the grade-6 conic–conic intersection blade.

TABLE 7. The meets of a conic $\mathbf{c}$ (grade 7). Meets of two objects whose grades sum to less than 8 vanish identically: low-grade point-objects are *joined*, not met.

$\mathbf{c} \vee B$	$\text{gr}(B)$	$\text{gr}(\mathbf{c} \vee B)$	meaning
conic $\vee$ conic	7	6	the 4 Bézout points (as $Q \wedge I_o^\flat$)
conic $\vee$ conic at ∞	5	4	incidence with the Veronese cone at ∞
conic $\vee$ quadpole	4	3	—
conic $\vee$ line at ∞ I_∞	3	2	the conic’s ideal dipole (Section 7.5)
conic $\vee$ twopole	2	1	point
conic $\vee$ point	1	0	incidence scalar, $q \vee \mathbf{c} = 0 \iff q \in \mathbf{c}$

9.2. The universal pipeline

For two conics $\mathbf{c}_1, \mathbf{c}_2$ in general position the meet is the grade-6 blade

$$I_6 = \mathbf{c}_1 \vee \mathbf{c}_2 \equiv p_1 \wedge p_2 \wedge p_3 \wedge p_4 \wedge I_o^\flat = Q \wedge I_o^\flat, \quad (44)$$

the *quadpole* Q of the four intersection points, gauge-fixed by the same $I_o^\flat$ that turns five points into a grade-7 conic — the intersection object is itself a rung of the $I_o^\flat$ -ladder of Section 5 (the $n = 4$ pencil blade). Incidence holds directly on the blade: $q \wedge I_6 = 0$ iff q is one of the four points. The gauge is stripped by the reciprocity mechanism of Section 6,

$$Q \equiv I_\infty^\flat \lrcorner I_6, \quad (45)$$

a pure grade-4 point object. What remains is to factor the quadpole into its four points — and this, by the design principle of Section 1, is exactly where the algebra hands over to scalar root extraction: splitting Q amounts to solving a quartic (a degenerate-conic pencil member, a resolvent cubic, then two $\pm\sqrt{}$ dipole splits — Ferrari’s solution in GA dress, cf. [10]). The four points come labelled by the Bézout reality split

$$r \text{ (finite real)} + m \text{ (finite conjugate)} + k \text{ (ideal)} = 4, \quad (46)$$

with each dipole’s reality read from its square as in Section 5.3. This route is valid for *every* pair of conics; the next subsection shows when it simplifies, and why.

9.3. Shared-factor / Bézout reduction

Each object of the taxonomy carries an *infinity signature* — its intersection with the line at infinity — fixed once and for all by a blade in its construction (Table 8). When two objects share ideal points by construction, that part of the Bézout budget is spent at infinity:

$$\# \text{ finite intersections} = 4 - \# \text{ forced common ideal points.} \quad (47)$$

TABLE 8. Infinity signatures: what each conic type meets I_∞ in, and the construction blade that forces it.

object	meets I_∞ in	forced by	reality
circle	the circular points I, J	$I_\infty^\flat$ (11)	imaginary
ellipse	its own ideal pair	weighted $I_\infty^\flat$	imaginary
hyperbola	its 2 asymptotic points	real ideal pair	real
parabola	1 double ideal point	$v_\infty \wedge v'_\infty$	real (tangent)
line (degen.)	<i>all</i> of I_∞	$I_\infty \equiv I_\infty^\flat \wedge e_\infty$	every ideal point

The table reads off the classical incidence facts as blade bookkeeping: two circles always share I, J , so at most 2 finite intersections remain (the radical-axis pair); a circle and a line share I, J as well ($I_\infty \supset I_\infty^\flat$), again 2 finite points; an ellipse and a hyperbola share nothing at infinity, so all 4 Bézout points are finite; two co-asymptotic hyperbolas share one real ideal point, leaving 3 finite ones; and two lines share the whole of I_∞ — an entire common component, so the finite meet is the single point of Table 7 read by incidence.

The round shortcut. When *both* objects contain the circular points — i.e. both belong to the circle/line (CGA) sub-family — two of the four Bézout points are exactly I, J , the quadpole factors as $Q \equiv D_2 \wedge I_\infty^\flat$ with D_2 a clean CGA point pair, and a second contraction strips the round gauge:

$$D_2 \equiv I_o^\flat \lrcorner (I_\infty^\flat \lrcorner I_6), \quad p_\pm = (D_2 \pm \sqrt{D_2^2}) (e_\infty \lrcorner D_2)^{-1}, \quad (48)$$

the classical conformal dipole extraction, with reality given by the sign of D_2^2 . We emphasise the validity boundary: the contraction in Equation (48) always *returns* a grade-2 object, but it is the true finite dipole only when both operands pass through I, J . For any other pair (an ellipse and a line, say) the two non-finite Bézout points are the conic’s *own* ideal pair, not the circular points; the $I_\infty^\flat$ -contraction then strips the wrong direction and the resulting grade-2 object is not the intersection. Outside the round sub-family, the universal pipeline of Section 9.2 is the only correct route. Table 9 summarises.

TABLE 9. Conic-family interactions. All meets have grade 6; the columns give the ideal points forced by Table 8, the resulting finite count (47), and the extraction route.

pair	forced ideal	finite	route
circle $\vee$ circle	I, J	2	shortcut (48)
circle $\vee$ line	I, J	2	shortcut (48)
circle $\vee$ ellipse	none	4	pipeline (Section 9.2)
ellipse $\vee$ ellipse / hyperbola	none	4	pipeline
ellipse $\vee$ line	ellipse's pair (imag.)	2	pipeline
hyperbola $\vee$ line	hyperbola's pair (real)	2	pipeline
parabola $\vee$ line	double ideal point	1	pipeline
line $\vee$ line	all of I_∞	1	incidence

Relation to the pencil method. The blade I_6 is also the algebraic carrier of the classical *pencil* of conics through the four points [14]: the conics $\lambda \mathbf{c}_1^* + \mu \mathbf{c}_2^*$ all contain the quadpole Q , and locating a degenerate member ($\Delta_3 = 0$, a cubic in $\lambda : \mu$ by Equation (34)) splits the intersection into line–conic problems. This is the route developed in [5], which the shared-factor organisation above complements: the pencil method computes the points; the infinity signatures predict their finite/ideal budget in advance and identify when the conformal shortcut applies, recovering the special configurations of [4] as instances of one rule.

10. Conclusion and Future Work

We have presented Conic Conformal Geometric Algebra as a single, consolidated foundation for two-dimensional conic geometry: one algebra ($\mathbb{R}_{5,3}$), one null working basis with a small set of special blades whose geometric roles — size, anisotropy, tilt, origin and infinity gauges — organise everything that follows. On this base we assembled the complete object taxonomy in both OPNS and IPNS form, structured by three construction ladders (bare multi-poles, the $I_o^\triangleright$ -gauged conic ladder, and the CGA sub-array embedded by $I_\infty^\triangleright$); closed-form, eigensolver-free extraction of every classical conic feature from the dual vector by inner products with the special blades; the explicit versors of the planar similarities together with a sharp delimitation of the versor group — similarities on the full conic family, conformal maps on the round sub-family only, and mere outermorphisms beyond; and a unified intersection calculus in which the regressive product, one gauge-stripping contraction, and an infinity-signature bookkeeping rule reduce every pairwise intersection to its Bézout budget. Throughout, the boundary between algebra and arithmetic is explicit and, we argue, optimal: every structural operation — construction, incidence, classification, feature extraction, intersection reduction — is geometric algebra, and the only numeric residue is root extraction, which carries the irreducible algebraic content of the problem.

Three directions remain open. First, the lift to three dimensions: CCGA is the planar restriction of QCGA ($\mathbb{R}_{9,6}$) [1], and most of the structure presented here — the gauge blades, the trace–deviatoric split, the shared-factor reduction — has a natural quadric analogue whose systematic development is future work. Second, a fully GA-native factorisation of degenerate conics: splitting a line pair into its two lines currently requires the adjugate of the coefficient matrix, and a blade-level factorisation would close the last coordinate step in the intersection pipeline. Third, the transformation group: shears and general projective maps act on CCGA only as outermorphisms, and whether a larger algebra can host them as versors — as $\mathbb{R}_{4,4}$ does for projective maps of points [12] — while retaining the conic object calculus is, to our knowledge, open.

References

- [1] Breuils, S., Fuchs, L., Hitzer, E., Nozick, V., Sugimoto, A.: Three-dimensional quadrics in extended conformal geometric algebras of higher dimensions from control points, implicit equations and axis alignment. *Advances in Applied Clifford Algebras* **29**(3) (2019). DOI 10.1007/s00006-019-0974-z. URL <https://hal.science/hal-02169419>. Publisher: Springer Verlag
- [2] Breuils, S., Nozick, V., Sugimoto, A., Hitzer, E.: Quadric conformal geometric algebra of $\mathbb{R}^{9,6}$. *Advances in Applied Clifford Algebras* **28**, 1–16 (2018)
- [3] Buchholz, S., Tachibana, K., Hitzer, E.M.S.: Optimal Learning Rates for Clifford Neurons. In: J.M. de Sá, L.A. Alexandre, W. Duch, D. Mandic (eds.) *Artificial Neural Networks – ICANN 2007*, Lecture Notes in Computer Science, pp. 864–873. Springer, Berlin, Heidelberg (2007). DOI 10.1007/978-3-540-74690-4_88
- [4] Byrtus, R., Derevianko, A., Vašík, P., Hildenbrand, D., Steinmetz, C.: On Specific Conic Intersections in GAC and Symbolic Calculations in GAALOP-Web. *Advances in Applied Clifford Algebras* **32**(1), 2 (2021). DOI 10.1007/s00006-021-01182-z. URL <https://doi.org/10.1007/s00006-021-01182-z>
- [5] Chomicki, C., Breuils, S., Biri, V., Nozick, V.: Intersection of conic sections using geometric algebra. In: *CGI ENGAGE 2023*. Shanghai, China (2023). URL <https://hal.science/hal-04210085>
- [6] De Keninck, S.: Non-parametric Realtime Rendering of Subspace Objects in Arbitrary Geometric Algebras. In: M. Gavrilova, J. Chang, N.M. Thalmann, E. Hitzer, H. Ishikawa (eds.) *Advances in Computer Graphics*, Lecture Notes in Computer Science, pp. 549–555. Springer International Publishing, Cham (2019). DOI 10.1007/978-3-030-22514-8_54
- [7] Dorst, L.: The Inner Products of Geometric Algebra. In: L. Dorst, C. Doran, J. Lasenby (eds.) *Applications of Geometric Algebra in Computer Science and Engineering*, pp. 35–46. Birkhäuser, Boston, MA (2002). DOI 10.1007/978-1-4612-0089-5_2. URL https://doi.org/10.1007/978-1-4612-0089-5_2
- [8] Dorst, L., Fontijne, D., Mann, S.: *Geometric Algebra for Computer Science: An Object-Oriented Approach to Geometry*. Elsevier (2010)

- [9] Easter, R.B., Hitzer, E.: Double Conformal Geometric Algebra. *Advances in Applied Clifford Algebras* **27**(3), 2175–2199 (2017). DOI 10.1007/s00006-017-0784-0. URL <https://doi.org/10.1007/s00006-017-0784-0>
- [10] Faucette, W.M.: A Geometric Interpretation of the Solution of the General Quartic Polynomial. *The American Mathematical Monthly* **103**(1), 51–57 (1996). DOI 10.2307/2975214. URL <https://www.jstor.org/stable/2975214>. Publisher: Mathematical Association of America
- [11] Fulton, W.: *Algebraic Curves: An Introduction to Algebraic Geometry*. Addison-Wesley Publishing Company, Advanced Book Program (1989)
- [12] Goldman, R., Mann, S.: $R(4, 4)$ As a Computational Framework for 3-Dimensional Computer Graphics. *Advances in Applied Clifford Algebras* **25**(1), 113–149 (2015). DOI 10.1007/s00006-014-0480-2. URL <https://doi.org/10.1007/s00006-014-0480-2>
- [13] Gregory, A.L., Lasenby, J., Agarwal, A.: The elastic theory of shells using geometric algebra. *Royal Society Open Science* **4**(3), 170065 (2017). DOI 10.1098/rsos.170065. URL <https://royalsocietypublishing.org/doi/10.1098/rsos.170065>. Publisher: Royal Society
- [14] Halsted, G.B.: *Synthetic projective geometry*. New York Wiley (1906). URL <http://archive.org/details/syntheticproject00halsuoft>
- [15] Hestenes, D.: The zitterbewegung interpretation of quantum mechanics. *Foundations of Physics* **20**, 1213–1232 (1990). DOI 10.1007/BF01889466
- [16] Hestenes, D.: *New Foundations for Classical Mechanics*. Springer Science & Business Media (2012)
- [17] Hrdina, J., Navrat, A., Vašík, P.: Geometric Algebra for Conics. *Advances in Applied Clifford Algebras* **28** (2018). DOI 10.1007/s00006-018-0879-2
- [18] Loučka, P., Vašík, P.: Algorithms for Conic Fitting Through Given Proper and Improper Waypoints in Geometric Algebra for Conics. *Advances in Applied Clifford Algebras* **34**(1), 6 (2024). DOI 10.1007/s00006-023-01308-5
- [19] Perwass, C.: *Geometric Algebra with Applications in Engineering*, vol. 4. Springer Berlin, Heidelberg (2009). DOI 10.1007/978-3-540-89068-3
- [20] Richter-Gebert, J.: *Perspectives on Projective Geometry: A Guided Tour Through Real and Complex Geometry*. Springer Science & Business Media (2011). Google-Books-ID: F_NP8Kub2XYC
- [21] Ruhe, D., Gupta, J.K., de Keninck, S., Welling, M., Brandstetter, J.: *Geometric Clifford Algebra Networks* (2023). DOI 10.48550/arXiv.2302.06594. URL <http://arxiv.org/abs/2302.06594>. ArXiv:2302.06594 [cs]

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